MTH 4110 — Differential Equations Study Notes

Separation of Variables

An ODE of the form $g(y)\,\dfrac{dy}{dx} = h(x)$ has the implicit solution given by $G(y) = H(x) + C$ where $G(y) = \int g(y)\,dy$ and $H(x) = \int h(x)\,dx$.

i.e. for $\dfrac{dy}{dx} = (3y-1)^2 \sin(2x)$:

$$\int (3y-1)^{-2}\,dy = \int \sin(2x)\,dx \implies \frac{-1}{3(3y-1)} = -\tfrac{1}{2}\cos(2x) + C.$$

Integrating Factor (Linear First-Order)

Consider the first-order linear ODE of the form $\dfrac{dy}{dx} + p(x)y = f(x)$. We define the integrating factor as $\mu(x) = e^{\int p(x)\,dx}$ and then multiply through by $\mu(x)$.

i.e. take $x\dfrac{dy}{dx} - 2y = x^3e^{3x}$, with $y(1) = e^3$. Rewrite as $\dfrac{dy}{dx} - \frac{2}{x}y = x^2e^{3x}$. Integrating factor: $\mu(x) = e^{\int -\frac{2}{x}\,dx} = e^{-2\ln x} = x^{-2}$. So:

$$x^{-2}\frac{dy}{dx} - 2x^{-3}y = e^{3x} \implies \tfrac{d}{dx}\left[x^{-2}y\right] = \tfrac{d}{dx}\left[\tfrac{1}{3}e^{3x}\right] \implies y = \frac{x^2e^{3x}}{3} + Cx^2.$$

Using $y(1) = e^3$: $C = \frac{2}{3}e^3$.

Integrating Factor for Nonexact Equations

Given a first-order ODE of the form $M(x,y) + N(x,y)y' = 0$, it is *exact* if there exists $\psi(x,y)$ such that $\psi_x = M$ and $\psi_y = N$. The previous ODE is exact if and only if $M_y = N_x$ for all $x, y \in \mathbb{R}$. If $M_y = N_x$, then $\psi(x,y) = \int M\,dx = F(x,y) + h(y)$. Take $\psi_y = \frac{d}{dy}\int M\,dx \overset{\text{set}}{=} N$. After solving for $h'(y)$, plug back into $\psi(x,y)$. Then the ODE becomes $\frac{d}{dx}[\psi(x,y)] = 0 \Rightarrow \psi(x,y) = C$.

If $M_y \neq N_x$, use the formula below and then use the result as the integrating factor:

- If $\dfrac{M_y - N_x}{N}$ is a function of only x, then $\mu(x) = e^{\int \frac{M_y - N_x}{N}\,dx}$.
- If $\dfrac{N_x - M_y}{M}$ is a function of only y, then $\mu(y) = e^{\int \frac{N_x - M_y}{M}\,dy}$.

i.e. for $6x^2y + (4x^3 + 15y^3)y' = 0$ with $y(2) = 1$: here $M = 6x^2y$, $N = 4x^3 + 15y^3$, so $M_y = 6x^2$, $N_x = 12x^2$. Since $\frac{N_x - M_y}{M} = \frac{6x^2}{6x^2y} = \frac{1}{y}$ depends only on y, the integrating factor is $\mu(y) = e^{\int \frac{1}{y}\,dy} = y$. Multiplying gives an exact equation; solving yields $2x^3y^2 + 3y^5 = 19$.

Characteristic Equation (Constant Coefficients)

If we have a linear homogeneous ODE with constant coefficients, we can look at the roots of the characteristic polynomial to determine the general solution. For $ay'' + by' + cy = 0$, the characteristic equation is $ar^2 + br + c = 0$.

- Two distinct real roots: $y(x) = c_1 e^{r_1 x} + c_2 e^{r_2 x}$.
- Repeated root: $y(x) = c_1 e^{rx} + c_2 x e^{rx}$.
- Complex roots $\alpha \pm \beta i$: $y(x) = e^{\alpha x}\big(c_1 \cos(\beta x) + c_2 \sin(\beta x)\big)$.

i.e. for $y''+4y'-12y = 0$ with $y(0) = 2$, $y'(0) = 12$: characteristic equation $r^2+4r-12 = (r+6)(r-2) = 0$, so $r = -6, 2$. General solution: $y(x) = c_1 e^{-6x} + c_2 e^{2x}$. Applying ICs gives $c_1 = -1$, $c_2 = 3$, so $y(x) = -e^{-6x} + 3e^{2x}$.

For higher-order ODEs, each root contributes according to its multiplicity. *i.e.* if the characteristic equation factors as $(r+1)^3(r^2-4r+5)(r^2+4)^2 = 0$, the roots are $r = -1$ (mult. 3), $r = 2 \pm i$ (each mult. 1), $r = \pm 2i$ (each mult. 2), giving order 9 and

$$y = c_1 e^{-x} + c_2 x e^{-x} + c_3 x^2 e^{-x} + c_4 e^{2x} \sin x + c_5 e^{2x} \cos x + c_6 \sin(2x) + c_7 x \sin(2x) + c_8 \cos(2x) + c_9 x \cos(2x).$$

Wronskian and Fundamental Sets

For y_1, y_2 on an interval I:

$$W[y_1, y_2](x) = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix} = y_1 y_2' - y_2 y_1'.$$

If $W \neq 0$ on I, then $\{y_1, y_2\}$ forms a fundamental set of solutions. Generalizes to n functions via an $n \times n$ determinant.

i.e. for three degree-1 polynomials $y_1 = Ax + B$, $y_2 = Cx + D$, $y_3 = Ex + F$, the third row of $W[y_1, y_2, y_3]$ is all zeros (since each $y_i'' = 0$), so $W = 0$ identically. Thus three linear functions cannot form a fundamental set for a third-order linear ODE.

Reduction of Order

Given one solution y_1 to $y'' + p(x)y' + q(x)y = 0$, look for a second solution of the form $y_2 = v(x)y_1(x)$. Letting $w = v'$, w satisfies the first-order linear ODE

$$y_1 w' + (2y_1' + p y_1) w = 0.$$

Solve for w, then $v = \int w\, dx$, and finally $y_2 = v y_1$.

i.e. for $x^2 y'' - 7xy' + 16y = 0$ on $x > 0$ with $y_1 = x^4$: standard form gives $p = -7/x$. Then

$$x^4 w' + \big(2(4x^3) - \tfrac{7}{x}(x^4)\big) w = 0 \;\Rightarrow\; x^4 w' + x^3 w = 0 \;\Rightarrow\; w = \tfrac{A}{x}.$$

So $v = A \ln x + B$. Choosing $A = 1$, $B = 0$ gives $y_2 = (\ln x) x^4$.

Undetermined Coefficients

For $ay'' + by' + cy = g(x)$ where g is a polynomial, exponential, sine/cosine, or product thereof, guess y_p of the same form with unknown coefficients. If any term in the guess overlaps with the complementary solution y_c, multiply by x^s where s is the smallest positive integer eliminating the overlap.

i.e. for $y''+y'-2y = xe^{4x}$: $r^2+r-2=(r+2)(r-1)=0$ gives $y_c = c_1e^{-2x}+c_2e^{x}$. No overlap with e^{4x}, so assume $y_p=(Ax+B)e^{4x}$. Computing y_p', y_p'' and substituting:

$$y_p''+y_p'-2y_p = 18Ax\,e^{4x}+(9A+18B)e^{4x} \overset{\text{set}}{=} xe^{4x}$$

gives $18A=1$, $9A+18B=0$, so $A=\frac{1}{18}$, $B=-\frac{1}{36}$. General solution: $y=(\frac{1}{18}x-\frac{1}{36})e^{4x}+c_1e^{-2x}+c_2e^{x}$.

Guess templates:

- $g(x)=$ polynomial of degree n: try $A_nx^n+\cdots+A_1x+A_0$.
- $g(x)=e^{\alpha x}$: try $Ae^{\alpha x}$.
- $g(x)=\sin(\beta x)$ or $\cos(\beta x)$: try $A\sin(\beta x)+B\cos(\beta x)$.
- Products and sums: multiply/add the corresponding guesses.

i.e. for $y^{(4)}-5y^{(3)}=2x^2$: characteristic equation $r^4-5r^3=r^3(r-5)=0$, so $y_c=c_1+c_2x+c_3x^2+c_4e^{5x}$. Since $1,x,x^2$ all appear in y_c, multiply the polynomial guess by x^3: $y_p=(Ax^2+Bx+C)x^3$.

Variation of Parameters

For $y''+p(x)y'+q(x)y=g(x)$ with $y_c=c_1y_1+c_2y_2$, look for $y_p=u_1(x)y_1+u_2(x)y_2$ where

$$u_1'=\frac{-y_2\,g}{W},\qquad u_2'=\frac{y_1\,g}{W},\qquad W=W[y_1,y_2].$$

Note g here is the RHS of the equation in *standard form* (coefficient of y'' equal to 1).

i.e. for $y''+16y=3\sec(4x)$: $r^2+16=0$ gives $r=\pm4i$, so $y_1=\sin(4x)$, $y_2=\cos(4x)$, $W=-4$. Then

$$u_1'=\frac{-\cos(4x)\cdot 3\sec(4x)}{-4}=\tfrac{3}{4}\;\Rightarrow\; u_1=\tfrac{3}{4}x,$$

$$u_2'=\frac{\sin(4x)\cdot 3\sec(4x)}{-4}=-\tfrac{3}{4}\tan(4x)\;\Rightarrow\; u_2=\tfrac{3}{16}\ln|\cos(4x)|.$$

General solution: $y=\frac{3}{4}x\sin(4x)+\frac{3}{16}\ln|\cos(4x)|\cos(4x)+c_1\sin(4x)+c_2\cos(4x)$.

Euler Equations

For $x^2y''+\alpha xy'+\beta y=0$ on $(0,\infty)$, try $y=x^r$. This gives the indicial equation $r(r-1)+\alpha r+\beta=0$.

- Distinct real r_1,r_2: $y=c_1x^{r_1}+c_2x^{r_2}$.
- Repeated root r: $y=c_1x^r+c_2x^r\ln x$.
- Complex roots $\alpha\pm\beta i$: $y=x^{\alpha}\big(c_1\cos(\beta\ln x)+c_2\sin(\beta\ln x)\big)$.

i.e. for $x^2y''-5xy'+10y=0$: $r(r-1)-5r+10=r^2-6r+10=0$ gives $r=3\pm i$, so $y=c_1x^3\sin(\ln x)+c_2x^3\cos(\ln x)$.

i.e. for $x^2y''+9xy'+16y=0$: $r(r-1)+9r+16=r^2+8r+16=(r+4)^2=0$ gives $r=-4$ (repeated), so $y=c_1x^{-4}+c_2x^{-4}\ln x$.

Series Solutions at Ordinary Points

If x_0 is an *ordinary point* of $y'' + p(x)y' + q(x)y = 0$ (meaning p, q are analytic at x_0), then there exist two linearly independent power-series solutions $y = \sum_{n=0}^{\infty} a_n (x - x_0)^n$. The radius of convergence is at least the distance from x_0 to the nearest singularity of p or q in the complex plane.

i.e. for $(x^2 + 16)y'' + xy' - y = 0$ at $x_0 = 3$: standard form has $\frac{x}{x^2+16}$ and $\frac{-1}{x^2+16}$ as coefficients, with singularities at $x = \pm 4i$. Distance from 3 to $4i$ is $\sqrt{9 + 16} = 5$, so the radius of convergence is at least 5.

Procedure. Substitute $y = \sum a_n x^n$, $y' = \sum n a_n x^{n-1}$, $y'' = \sum n(n-1) a_n x^{n-2}$, reindex so all sums use the same power of x, combine into a single sum, and set the coefficient of each x^n to zero to get a recurrence relation.

i.e. for $y'' + 3xy' - y = 0$:

$$\sum_{n=2}^{\infty} n(n-1) a_n x^{n-2} + 3 \sum_{n=1}^{\infty} n a_n x^n - \sum_{n=0}^{\infty} a_n x^n = 0.$$

Reindexing the first sum and combining: $2a_2 - a_0 + \sum_{n=1}^{\infty} \big[(n+2)(n+1)a_{n+2} + (3n-1)a_n\big] x^n = 0$. So $a_2 = a_0/2$ and

$$a_{n+2} = \frac{-(3n-1)}{(n+2)(n+1)}\, a_n, \qquad n \geq 1.$$

This gives a general solution $y = a_0 y_1(x) + a_1 y_2(x)$ with $R = \infty$.

Shortcut without sigma notation. If $y(x)$ is unknown and the coefficients of the ODE are themselves functions like e^x, expand everything as power series and match coefficients term by term.

i.e. for $y'' + e^x y = 0$ with $y = a_0 + a_1 x + a_2 x^2 + a_3 x^3 + \cdots$ and $y'' = 2a_2 + 6a_3 x + 12 a_4 x^2 + \cdots$:

$$(2a_2 + 6a_3 x + \cdots) + \big(1 + x + \tfrac{x^2}{2!} + \cdots\big)(a_0 + a_1 x + a_2 x^2 + \cdots) = 0.$$

Matching the constant term: $2a_2 + a_0 = 0 \Rightarrow a_2 = -a_0/2$. Matching x: $6a_3 + a_1 + a_0 = 0 \Rightarrow a_3 = (-a_1 - a_0)/6$.

Frobenius / Regular Singular Points

A point $x_0 = 0$ is a *regular singular point* of $y'' + p(x)y' + q(x)y = 0$ if $xp(x)$ and $x^2 q(x)$ are analytic at 0. Let $p_0 = \lim_{x \to 0} xp(x)$ and $q_0 = \lim_{x \to 0} x^2 q(x)$. Then:

- **Corresponding Euler equation:** $x^2 y'' + p_0 x y' + q_0 y = 0$.
- **Indicial equation:** $r(r-1) + p_0 r + q_0 = 0$. Its roots are the *exponents of the singularity.*

Look for solutions of the form $y = \sum_{n=0}^{\infty} a_n x^{n+r}$.

i.e. for $2x^2 y'' - (3x + x^7)y' + (5\cos^3 x)y = 0$ on $x > 0$: standard form gives $p(x) = -\frac{3x+x^7}{2x^2}$, $q(x) = \frac{5\cos^3 x}{2x^2}$. Then $p_0 = \lim_{x\to 0} xp(x) = -3/2$ and $q_0 = \lim_{x \to 0} x^2 q(x) = 5/2$. The corresponding Euler equation is $x^2 y'' - \frac{3}{2} x y' + \frac{5}{2} y = 0$, the indicial equation is $r(r-1) - \frac{3}{2} r + \frac{5}{2} = 0$, i.e. $2r^2 - 5r + 5 = 0$, giving exponents $r = \frac{5}{4} \pm \frac{i\sqrt{15}}{4}$.

i.e. for $4x^2 y'' + (1 + x)y = 0$ on $x > 0$: substituting $y = \sum a_n x^{n+r}$ and matching the lowest power gives the indicial equation $4r^2 - 4r + 1 = 0$, so $r = 1/2$ (double root). The recurrence becomes

$$a_n = \frac{-a_{n-1}}{(2n+1)(2n-1) + 1}, \qquad n \geq 1.$$

First terms: $a_1 = -a_0/4$, $a_2 = a_0/(16 \cdot 4)$, $a_3 = -a_0/(36 \cdot 16 \cdot 4)$, so

$$y(x) = \sqrt{x}\left(1 - \tfrac{1}{4}x + \tfrac{1}{64}x^2 - \tfrac{1}{2304}x^3 + \cdots\right).$$

Laplace Transforms — Definition

$$\mathcal{L}\{f(t)\} = \int_0^\infty e^{-st} f(t)\, dt$$

provided the integral converges.

i.e. for $f(t) = 2e^{-7t}$:

$$\mathcal{L}\{2e^{-7t}\} = \int_0^\infty 2e^{-st}e^{-7t}\, dt = 2\int_0^\infty e^{-t(s+7)}\, dt = \frac{2}{-(s+7)}e^{-t(s+7)}\Big|_0^\infty = \frac{2}{s+7}, \quad s > -7.$$

Common Laplace Transforms

For any real a, positive integer n, $c > 0$, and $t_0 \geq 0$:

$$\mathcal{L}\{1\} = \tfrac{1}{s},\ s > 0 \qquad \mathcal{L}\{t^n\} = \tfrac{n!}{s^{n+1}},\ s > 0$$
$$\mathcal{L}\{e^{at}\} = \tfrac{1}{s-a},\ s > a \qquad \mathcal{L}\{t^n e^{at}\} = \tfrac{n!}{(s-a)^{n+1}},\ s > a$$
$$\mathcal{L}\{\sin(at)\} = \tfrac{a}{s^2+a^2},\ s > 0 \qquad \mathcal{L}\{\cos(at)\} = \tfrac{s}{s^2+a^2},\ s > 0$$
$$\mathcal{L}\{u_c(t)\} = \tfrac{e^{-sc}}{s},\ s > 0 \qquad \mathcal{L}\{\delta(t-t_0)\} = e^{-st_0},\ s > 0$$

Translation theorems. Let $F(s) = \mathcal{L}\{f(t)\}$. For $c > 0$:

- $\mathcal{L}\{e^{ct}f(t)\} = F(s-c)$.
- $\mathcal{L}\{u_c(t)f(t-c)\} = e^{-cs}F(s)$.

Derivative formulas. $\mathcal{L}\{y'\} = sY - y(0)$, $\mathcal{L}\{y''\} = s^2Y - sy(0) - y'(0)$.

Solving IVPs with Laplace Transforms

Take the Laplace transform of both sides, solve algebraically for $Y(s)$, decompose into known transforms (often via partial fractions), and invert.

i.e. for $y'' - 2y' - 3y = 6e^{4t}$, $y(0) = -3$, $y'(0) = 1$:

$$(s^2Y + 3s - 1) - 2(sY + 3) - 3Y = \tfrac{6}{s-4} \implies (s^2 - 2s - 3)Y = -3s + 7 + \tfrac{6}{s-4}.$$

So $Y = \dfrac{-3s^2 + 19s - 22}{(s-3)(s+1)(s-4)}$. Partial fractions: $\dfrac{A}{s-3} + \dfrac{B}{s+1} + \dfrac{C}{s-4}$ with $A = -2$, $B = -11/5$, $C = 6/5$. Inverting:

$$y(t) = -2e^{3t} - \tfrac{11}{5}e^{-t} + \tfrac{6}{5}e^{4t}.$$

i.e. to find $\mathcal{L}^{-1}\left\{\dfrac{s-2}{s^2+9} + \dfrac{3}{s^5}\right\}$: split as

$$\mathcal{L}^{-1}\left\{\tfrac{s}{s^2+9}\right\} - \tfrac{2}{3}\mathcal{L}^{-1}\left\{\tfrac{3}{s^2+9}\right\} + \tfrac{3}{4!}\mathcal{L}^{-1}\left\{\tfrac{4!}{s^5}\right\} = \cos(3t) - \tfrac{2}{3}\sin(3t) + \tfrac{3}{4!}t^4.$$

Autonomous Equations and Stability

An equation $\dfrac{dy}{dx} = f(y)$ is autonomous. The *critical points* (equilibrium solutions) are roots of $f(y) = 0$. To classify, examine the sign of f on either side of each critical point:

- Sign goes $+ \to -$: *stable* (asymptotically attracting).
- Sign goes $- \to +$: *unstable* (repelling).
- Same sign on both sides: *semi-stable*.

i.e. for $\dfrac{dy}{dx} = -3y(y^2 - 7y + 10) = -3y(y-5)(y-2)$: critical points are $y = 0, 2, 5$. Testing signs: $y = 0$ and $y = 5$ are stable; $y = 2$ is unstable.

Existence and Uniqueness (Linear First-Order)

For the linear IVP $\dfrac{dy}{dx} + p(x)y = g(x)$, $y(x_0) = y_0$: the largest interval of existence and uniqueness is the largest interval containing x_0 on which both p and g are continuous.

i.e. for $\dfrac{dy}{dx} - \frac{5}{\sin^3 x}y = \frac{6}{x-2}$, $y(1) = 2$: $-5/\sin^3 x$ is discontinuous at $x = n\pi$ ($n \in \mathbb{Z}$), and $6/(x-2)$ is discontinuous at $x = 2$. The largest interval containing $x_0 = 1$ on which both are continuous is $(0, 2)$.